

THE REGULATION OF “DOPAMINE-HACKING,” AN ABUSE OF BIOLOGICAL MECHANISMS BY COMPANIES AND SOFTWARE ENGINEERS ALIKE

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TESTIMONIES FROM OUR FELLOW CLASSMATES HIGHLIGHT HOW THIS “BRAIN-HIJACKING” STRUGGLE AFFECTS THEIR LIVES

- Our peer, Rosalynd Van Pelt, said this: “When I had difficult papers to write, I lost motivation and used scrolling as a way to forget it. Even if the app had an automated suggested break, my attention was already too low to consider my present moment.”
- Another classmate of ours, Marlee Wilson, was kind enough to share how they also have been personally impacted, saying, “While I can't recall a specific instance of this, I know it has happened to me multiple times. It makes me so sad to think that I am not being aware or hospitable to people that I care about when I'm on my phone!”



CREDIBILITY & THESIS



Main Point #1

There are numerous mechanisms behind how technology companies and software developers intentionally exploit the dopamine systems of our neurobiology, which keep us engaged with their product or provided service.

- Subpoint #1: One primary method of “dopamine-hijacking” is utilizing it to lock users into an “infinite scroll” (better known to some of you as the “doom scroll”) so that their product becomes your primary focus, and you forget all your pressing responsibilities
 - Sub-Subpoint: Software engineers utilize “infinite scroll” to remove natural stopping points, keeping users in a state of constant consumption.
 - Sub-Subpoint: This design choice prevents the brain from processing the “completion” of a task, leading to the “brain fry” many of us feel.



Main Point #1

There are numerous mechanisms behind how technology companies and software developers intentionally exploit the dopamine systems of our neurobiology, which keep us engaged with their product or provided service.

- Subpoint #2: Another primary method of dopamine-hijacking is the use of variable-ratio reinforcement.
 - Sub-Subpoint: According to research by Montag et al. (2019), these features are specifically designed to trigger a physiological addiction response rather than a choice.
 - Sub-Subpoint: This tactic targets internal reward circuits, making the "addiction" a result of intentional engineering.



Main Point #1

There are numerous mechanisms behind how technology companies and software developers intentionally exploit the dopamine systems of our neurobiology, which keep us engaged with their product or provided service.

- Subpoint #3: Dopamine-hijacking's impact on student well-being.
 - Sub-Subpoint: This exploitation leads to significant sleep problems and academic struggles for students in our generation (Center for Humane Technology, 2023).
 - Sub-Subpoint: My fellow classmates are essentially the "victims" of these strategies, facing daily battles with their own focus.



Main Point #2

We must shift from "self-control" to policy-based solutions

- Subpoint #1: We must address the counterargument that distraction is merely a "failure of willpower."
 - Sub-Subpoint: A common belief is that distraction is a personal failure, but neuroimaging evidence shows that these apps actually suppress the prefrontal cortex (Lorenz, R. C., et al. (2015)).
 - Sub-Subpoint: Because this area of the brain is required for executive function, users are biologically disadvantaged when trying to resist.



Main Point #2

We must shift from "self-control" to policy-based solutions

- Subpoint #2: Benefits and procedure for implementing neuro-ethical design standards.
 - Sub-Subpoint: We must support legislative policies, like the SMART Act, to limit exploitative features used in software engineering (SMART Act, S.2314, 116th Cong. (2019)).
 - Sub-Subpoint: This act aims to ban the very features, such as auto-play and infinite scroll, that create the dopamine loops we've discussed.



Main Point #2

We must shift from "self-control" to policy-based solutions

- Subpoint #3: Shifting the burden of responsibility from the individual onto the policy-makers and developers
 - Sub-Subpoint: My goal is to pursue a career where I serve people and protect their health; this starts with advocating for neuro-ethics (Bhargava, V. R., & Velasquez, M. (2021)).
 - Sub-Subpoint: Policy restores cognitive autonomy by ensuring companies can no longer operate outside the confines of law.



Main Point #3

What life looks like when we reclaim our cognitive autonomy.

- Subpoint #1: Imagine what life could be like with you back in the "driver's seat."
 - Sub-Subpoint: Imagine the mental clarity you would gain by being in complete control of your own mind and actions.
 - Sub-Subpoint: This clarity allows for greater academic success and more meaningful professional relationships.



Main Point #3

What life looks like when we reclaim our cognitive autonomy.

- Subpoint #2: Let's hear an inspiring transformation story from somebody who has reformed their technology use.
 - Sub-Subpoint: Consider a student who, after breaking free from these dopamine loops, regained their focus and peace of mind.
 - Sub-Subpoint: This transformation shows that we deserve to be in control of our own cognitive health.



Main Point #3

What life looks like when we reclaim our cognitive autonomy.

- Subpoint #3: Cognitive autonomy will help protect your future careers and well-being.
 - Sub-Subpoint: As future professionals, we must ensure our focus is preserved to be successful in our chosen fields.
 - Sub-Subpoint: Reclaiming our autonomy means technology becomes a tool we use, rather than a system that uses us.



CONCLUSION

- **Summary:** (Restate goal and main points): We have examined the science of exploitation, why willpower is insufficient against engineering, and how the SMART Act can restore our autonomy. You can take action yourself by supporting neuro-ethical legislation and advocating for more standards that protect our collective cognitive health. Give your local congressmen a call, they'll never make a change if they don't know the extent of this issues negative affects.
- **Memorable Close:** (Provide closure and leave a lasting impression): To conclude, I want to leave you with a comparison: casinos are designed to turn off executive-functioning regions of your brain through the intentional lack of windows and clocks to keep you gambling. Technology should be a tool that we pick up and put down, not a casino with no clocks. We must not let ourselves become enslaved by it, forgetting our responsibilities...forgetting ourselves. Let's all advocate for a world where the exit door is easy to find and reclaim control over our lives!



REFERENCES

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